

NANO MATERIALS

Introduction

All materials are composed of grains, which in turn comprise many atoms. These grains are usually invisible to the naked eye, depending on their size. Conventional materials have grains varying in size anywhere from 100's of microns (μm) to millimeters (mm). A micron (μm) is a micrometer or a millionth (10^{-6}) of a meter. An average human hair is about 100 μm in diameter. A nanometer (nm) is even smaller a dimension than a μm and is a billionth (10^{-9}) of a meter. A nanocrystalline material has grains on the order of 1-100 nm. The average size of an atom is on the order of 1 to 2 angstroms ($\text{\AA}$) in radius.

1 nanometer comprises 10 $\text{\AA}$, and hence in one nm, there may be 3-5 atoms, depending on the atomic radii. Nanocrystalline materials are exceptionally strong, hard, and ductile at high temperatures, wear-resistant, corrosion-resistant, and chemically very active. Nanocrystalline materials, or Nanomaterials, are also much more formable than their conventional, commercially available counterparts.

5.4.2 Definitions

1. Nanotechnology

Nanotechnology is a field of applied science and technology which deals with the matter on the atomic and molecular scale, normally 1 to 100 nanometers, and the fabrication of devices with critical dimensions that lie within that size range.

2. Nanomaterials

Nano materials are the materials with grain sizes of the order of nano meter (10^{-9} m) i.e., (1-100 nm). It may be a metal, alloy, inter metallic (or) ceramic.

Nanomaterials are the materials with atoms arranged in nano sized clusters which become the building block of the material. Any Material with a size between 1 and 100 nm [10^{-9} m to 10^{-7} m] is also called Nanomaterials.

5.4.3 Synthesis of Nanomaterials

The Nano materials can be synthesized by two processes, they are

1. Top – down approach
2. Bottom – up approach

1. Top – down approach

The removal or division of bulk material or the miniaturization of bulk fabricative processes to produce the desired nanostructure is known as top-down approach. It is the process of breaking down bulk material to Nano size.

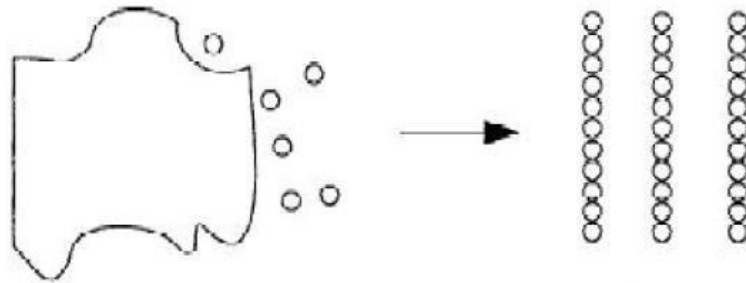


Fig.5.4 Synthesis of Nanomaterials for Top – down approach

Types of Top – down Methods

- a. Milling
- b. Lithographic
- c. Machining

2. Bottom – up approach

Molecules and even nano particles can be used as the building block for producing complex nanostructures. This is known as Bottom – up approach. The Nano particles are made by building atom by atom.

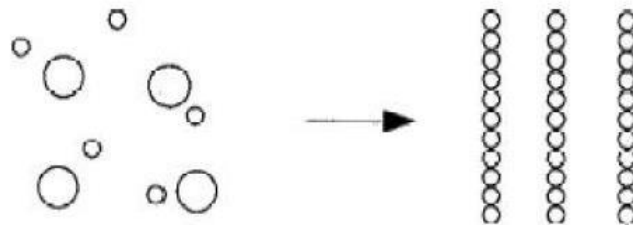


Fig.5.5 Synthesis of Nanomaterials for Bottom – up approach

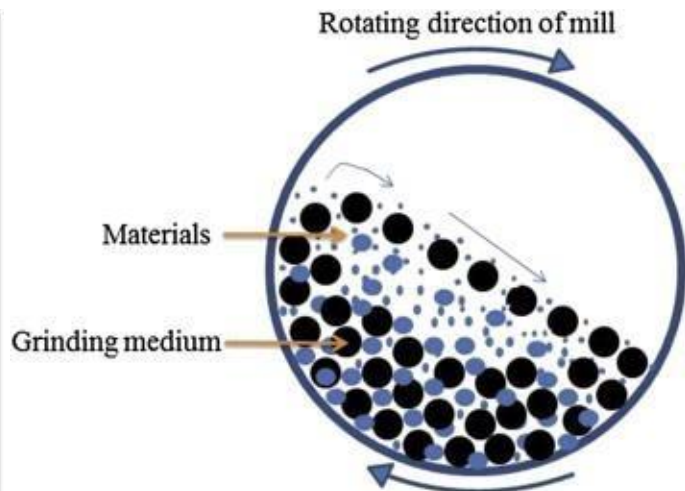
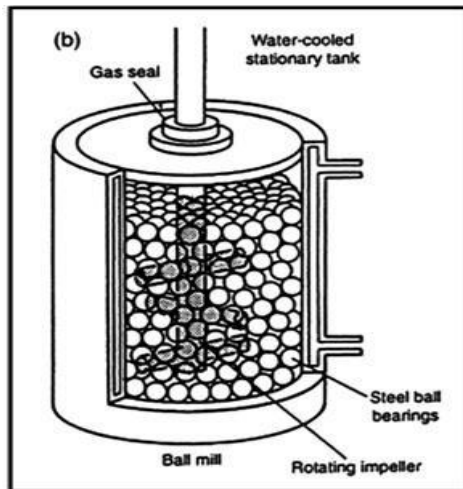
Types of Bottom up Methods

- a) Vapour phase deposition Method
- b) Molecular beam epitaxy Method
- c) Plasma assisted deposition Method
- d) Metal Organic Vapour Phase Epitaxy [MOVPE]
- e) Liquid phase process [Colloidal method and Sol – Gel method]

BALL MILLING METHOD:

Steps in ball milling method:

1. The ball milling method consists of balls and a mill chamber. Therefore over all a ball mill contains a stainless steel container and many small iron, hardened steel, silicon carbide, or tungsten carbide balls are made to rotate inside a mill (drum).
2. The powder of a material is taken inside the steel container. This powder will be made into nano size using the ball milling technique. A magnet is placed outside the container to provide the pulling force to the material and this magnetic force increases the milling energy when milling container or chamber rotates the metal balls.



3. The ball to material mass ratio is normally maintained at 2 ratio1.
4. These silicon carbide balls provide very large amount of energy to the material powder and the powder then get crushed. This process of ball milling is done approximately 100 to 150 hrs to get uniform fine powder.
5. Ball milling is a mechanical process and thus all the structural and chemical changes are produced by mechanical energy.

Advantages of ball milling process:

1. Nanopowders of 2 to 20 nm in size can be produced. The size of nanopowder also depends upon the speed of the rotation of the balls.
2. It is an inexpensive and easy process.

Disadvantages;

1. As the process is not so sophisticated, therefore the shape of the nanomaterial is irregular.
2. There may be contaminants inserted from ball and milling additives.
3. This method produces crystal defects.

VAPOUR PHASE DEPOSITION:

In general, PVD is used to deposit titanium, titanium nitrate, tantalum, tantalum nitrate, aluminum and a very thin film of copper called seed layer. CVD is used to deposit tungsten, titanium, again copper seed layer, silicon di-oxide, silicon nitride, etc. Electrochemical deposition is used for depositing copper. Spin-on coating is used to deposit insulators or dielectric materials, which are usually organic in nature.

Historically, some of the metals were evaporated and then the vapor was deposited onto the wafer. The wafer would be kept at lower temperature and the vapor will cool down and form the solid metal. These methods used large amount of energy and had other difficulties. It could not be used for all materials, and particularly for alloys. The material will also deposit on other places like the chamber walls. Hence, it is not currently used for IC manufacturing.

If the raw material is taken in gaseous form and the final material is deposited on the wafer after reaction, it is called chemical vapor deposition (CVD). If the material is taken in solid state, but sent as very small particles or atoms and deposited on the wafer without any reaction, it is called physical vapor deposition (PVD).

PHYSICAL VAPOUR DEPOSITION (PVD)

Physical Vapour Deposition (PVD) is a collective set of processes used to deposit thin layers of material, typically in the range of few nanometers to several micrometers. PVD processes are environmentally friendly vacuum deposition techniques consisting of three fundamental steps.

- Vaporization of the material from a solid source assisted by high temperature vacuum or gaseous plasma.
- Transportation of the vapor in vacuum or partial vacuum to the substrate surface.
- Condensation onto the substrate to generate thin films.

Different PVD technologies utilize the same three fundamental steps but differ in the methods used to generate and deposit material. The two most common PVD processes are thermal evaporation and sputtering. Thermal evaporation is a deposition technique that relies on vaporization of source material by heating the material using appropriate methods in vacuum. Sputtering is a plasma-assisted technique that creates a vapor from the source target through bombardment with accelerated gaseous ions (typically Argon). In both evaporation and sputtering, the resulting vapor phase is subsequently deposited onto the desired substrate through a condensation mechanism. Deposited films can span a range of chemical compositions based on the source material(s). Further compositions are accessible through reactive deposition processes. Relevant examples include co-deposition from multiple sources, reaction during the transportation stage by introducing a reactive gas (nitrogen, oxygen or simple hydrocarbon containing the desired reactant), and post-deposition modification through thermal or mechanical processing. PVD is used in a variety of applications, including fabrication of microelectronic devices, interconnects, battery and fuel cell electrodes, diffusion barriers, optical and conductive coatings, and surface modifications.

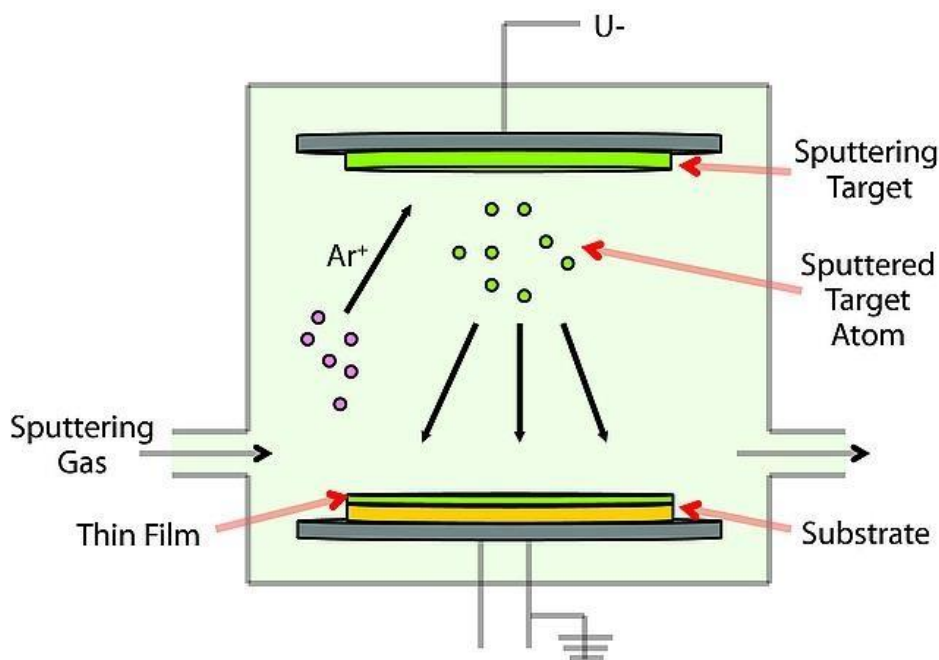


Figure: Schematic of a PVD chamber

PVD (Physical Vapour Deposition)

Advantages:

- ✓ No environmentally damaging materials and emissions, no toxic reaction products
- ✓ Great variety of coatings can be produced
- ✓ Coating temperature below the final heat treatment temperature of most steels
- ✓ Small, precisely reproducible coating thickness (accurate surface replication, true to size,
- ✓ High wear resistance
- ✓ Low frictional coefficient

Disadvantages:

- ✓ Bore holes, slots etc. can only be coated down to a depth equal to the diameter or width of the opening
- ✓ Corrosion resistant only under certain conditions
- ✓ In order to achieve a uniform coating thickness, the parts to be coated must be rotated during processing

CHEMICAL VAPOUR DEPOSITION (CVD)

A large CVD chamber may be of 6 ft long and 3 ft diameter. In this, a material may be deposited on a lot (i.e. a batch) of 25 wafers. The CVD chamber will have pipes and automatic valves to create vacuum and to supply desired gases at controlled rates. The material to be deposited (e.g. tungsten) will be in the form of a compound, in the gaseous form. The CVD chamber will also have facilities to heat the wafers in a controlled fashion. A simple schematic of a CVD chamber is given in Figure.

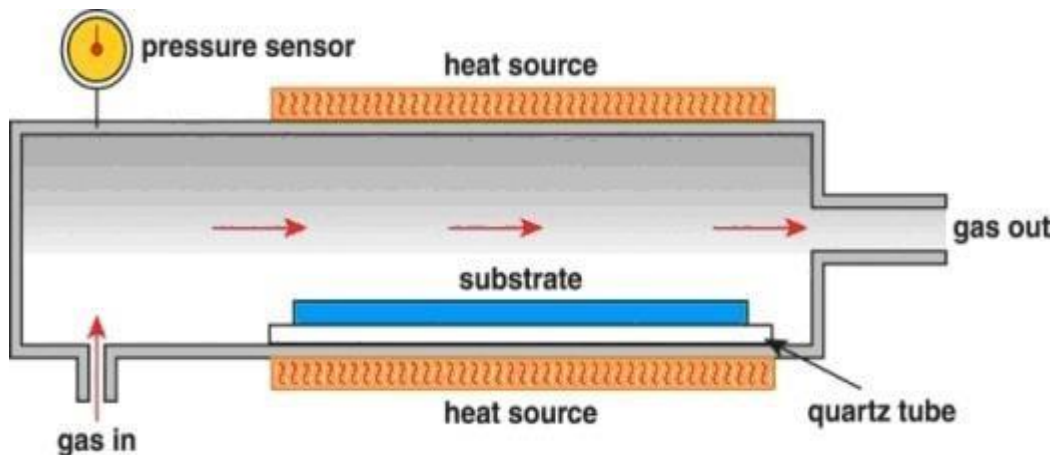


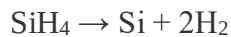
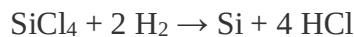
Figure: Schematic of a CVD chamber

Operation: First the wafers are kept in the chamber and the chamber is evacuated. Then the wafers will be heated to the desired temperature. Next the gases are supplied and they react only on the surface of the wafer and deposit the material. For example, let us assume that we want to deposit tungsten on the wafer. For this, a mixture of tungsten hexa fluoride, hydrogen and nitrogen (all in gaseous form) will be supplied to the chamber at low pressure. The temperature of the wafer would be maintained between 150 to 300 °C. In the CVD technology, the phrase “high temperature” refers to 800+ °C and low temperature may refer to “less than 400 °C. Very low temperature may refer to room temperature.

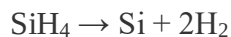
In most of the CVD tools, all the chamber walls will be maintained at relatively lower temperature compared to the wafer. Only the wafers would be heated. WF_6 and H_2 will react only at high temperature. Hence the reaction will occur only on the wafer and not on the chamber walls. This type of CVD process is called low pressure CVD or LPCVD. Using this method many materials such as W, Ti, TiN, Cu, SiO_2 , Si_3N_4 , Si etc can be deposited on wafer. The side wall coverage will also be good.

Some example reactions to deposit various materials are given below.

1. To deposit silicon



2. To deposit poly silicon (hyper pure form of silicon)



CVD (Chemical Vapour Deposition)

Advantages:

- ✓ High wear resistance
- ✓ Economic production of thicker coatings
- ✓ Also suitable for bore holes, slots etc.

Disadvantages:

- ✓ High processing temperatures
- ✓ Coatings with several metals (e.g. TiAlN) are not possible
- ✓ Edges become rounded (coating thickness)
- ✓ Uses ecologically problematic, toxic metal chlorides

PROPERTIES OF THE NANOPARTICLES

Magnetic properties:

1. Magnetic nanoparticles are used in a range of applications like imaging, bioprocessing, refrigeration as well as high storage density magnetic memory media.
2. The large surface area to volume ratio results in a substantial proportion of atoms having different magnetic coupling with neighboring atoms leading to differing magnetic properties.
3. Bulk gold and platinum are non magnetic but at the nano size they act as magnetic particles. Au nanoparticles become ferromagnetic when they are capped with the appropriate molecules such as thiol.
4. Giant magnetoresistance (GMR) is a phenomenon observed in nanoscale multilayers consisting of strong ferromagnet (Fe, Co, Ni) and a weaker magnetic or non magnetic buffer (Cr, Cu). It is usually employed in data storage and sensing.

Optical properties:

In small nano clusters the effect of reduced dimensionality on electronic structure has the most profound effect on the energies of highest occupied molecular orbital (HOMO) which is valence band and the lowest unoccupied molecular orbital (LUMO), essentially the conduction band.

1. The optical emission and adsorption occurs when the transition of the electrons occur between these two states.
2. Semiconductors and many metals show large changes in optical properties such as color, as a function of particle size.
3. Colloidal suspensions of gold nano particles have a deep red color which becomes progressively more yellow as the particle size increases.
 - Gold spheres of 10-20nm exhibit red color
 - Gold spheres of 2-5nm exhibit yellow color.
 - Gold spheres of >20nm exhibit purple colorSimilarly,
 - Silver particles of 40nm exhibit blue color

- Silver particles of 100nm exhibit yellow color
- Prism shaped Silver particles red color.

Other properties which may be affected by reduced dimensionality include photocatalysis, photoconductivity, photoemission and electroluminescence.

Electronic properties:

1. The changes which occur in electronic properties as the system length scale is reduced are related mainly to the increasing influence of the wave-like property of the electrons (quantum mechanical effects) and the scarcity of scattering centres.
2. As the size of the system becomes comparable with the de Broglie wavelength of the electrons, the discrete nature of the energy states becomes apparent once again, although a fully discrete energy spectrum is only observed in systems that are confined in all three dimensions.
3. In certain cases, conducting materials become insulators below a critical length scale, as the energy bands cease to overlap. Owing to their intrinsic wave-like nature, electrons can tunnel quantum mechanically between two closely adjacent nanostructures, and if a voltage is applied between two nanostructures which aligns the discrete energy levels in the DOS, resonant tunnelling occurs, which abruptly increases the tunnelling current.
4. Conduction in highly confined structures, such as quantum dots, is very sensitive to the presence of other charge carriers and hence the charge state of the dot.
5. These Coulomb blockade effects result in conduction processes involving single electrons and as a result they require only a small amount of energy to operate a switch, transistor or memory element.
6. All these phenomena can be utilized to produce radically different types of components for electronic, optoelectronic and information processing applications, such as resonant tunnelling transistors and single-electron transistors.

APPLICATIONS OF NANO PHASE MATERIALS:

1. Mechanical Engineering

- a) Since they are stronger, lighter etc., they are used to make hard metals.
- b) Smart magnetic fluids are used in vacuum seals, magnetic separators etc.
- c) They are also used in Giant Magneto Resistant (GMR) spin valves.
- d) Nano- MEMS (Micro-Electro Mechanical Systems) are used in ICs, optical switches, pressure sensors, mass sensors etc.

2. Electrical, Electronics and Communication Engineering

- a) Orderly assembled nanomaterials are used as quantum electronic devices and photonic crystals.
- b) Some of the nanomaterials are used as sensing elements. Especially the molecular nanomaterials are used to design the robots, assemblers etc.
- c) They are used in energy storage devices such as hydrogen storage devices, magnetic refrigeration and in ionic batteries.
- d) Dispersed nanomaterials are used in magnetic recording devices, rocket propellant, solar cells, fuel cells, etc.
- e) Recently nano-robots were designed, which are used to remove the damaged cancer cells and also to modify the neutron network in human body.

3. Computer Science Engineering and IT

- a) Nano-materials are used to make CD's and semiconductor laser.
- b) These materials are used to store the information in smaller chips.
- c) They are used in mobiles, lap – tops etc
- d) Further they are used in chemical / Optical computers.
- e) Nano – dimensional photonic crystals and quantum electronic devices plays a vital role in the recently developed computers.

4. Bio-Medical and Chemical Engineering

- a) Consolidated state nanoparticles are used as catalyst, electrodes in solar and fuel cells.
- b) Bio-sensitive nanoparticles are used in the production of DNA –chips, bio-sensors etc.
- c) Nano-structed ceramic materials are used in synthetic bones.
- d) Few nanomaterials are also used in adsorbents, self –cleaning glass, fuel additives, drugs, ferrofluids, paints etc.
- e) Nano –metallic colloids are used as film precursors.

Carbon Nanotubes (CNTs)

What are Carbon nanotubes (CNT)? Mention different types of CNTs.

Carbon nanotubes are a graphite sheet consisting of covalently bonded carbon atoms in hexagonal type arrangement. The sheet is rolled up into a cylinder with the ends closed by hemispherical graphite domes. Carbon nanotubes can be divided into two basic classes namely

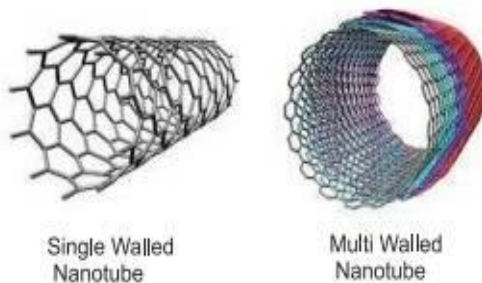


Figure 8.1: SWNT and MWNT

Single (SWNT) and Multi-wall (MWNT) nanotubes

SWNT

A single atomic layer forms the wall.
Tube diameter of between 0.6 and 5 nm.
Well characterized structure and properties

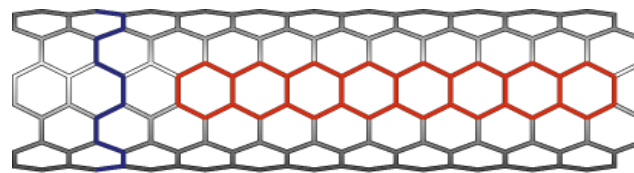
MWNT

Inner diameter: 1.5 – 15 nm
Outer diameter: 2.5 – 50 nm
~50 layers
Many structural defects.

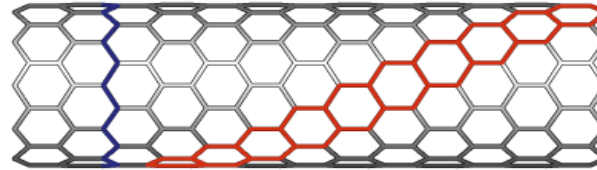
1. Single wall nanotubes (SWNT)
2. Multi wall nanotubes (MWNT)

SWNT's are formed from the single graphite layer with typical diameter in the range 0.4 to 2 nm. MWNT's include structures formed in coaxial arrangement of several (2-100) graphite cylinders and their external diameter ranges from 10 to 100 nm. Based on the chirality CNT structures are classified into 3 types. Namely,

1. Zig-zag structure
2. Armchair structure
3. Chiral structure



Armchair



Zig-zag

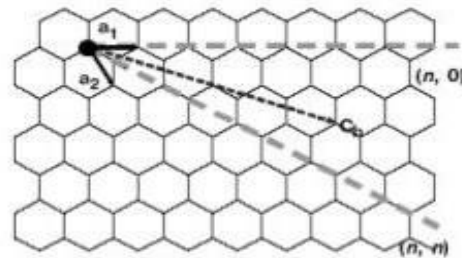
Structure

The graphene sheet can be rolled in different ways to get the three type of SWNTS:

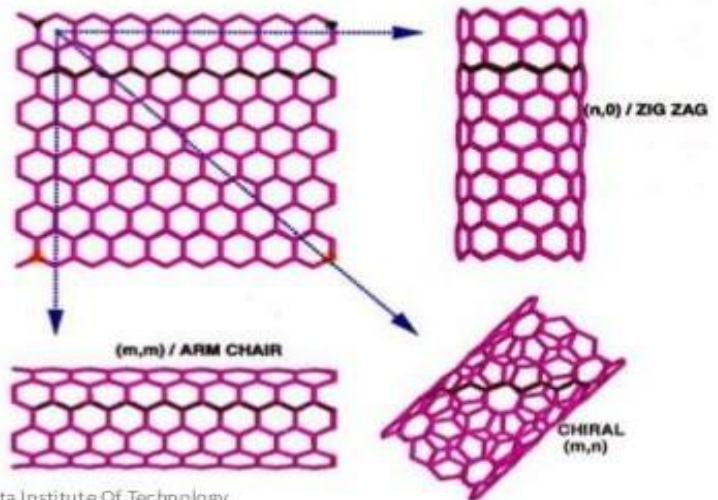
1. Zigzag
2. Armchair
3. Chiral

These three types are classified on the Chiral vector.

1. Zigzag
In these the graphene sheet is rolled up along a vector greater than the chiral angle.
2. Armchair
In these the graphene sheet is rolled up along the vector smaller than the chiral angle.
3. Chiral
In these it is rolled up on the chiral vector.



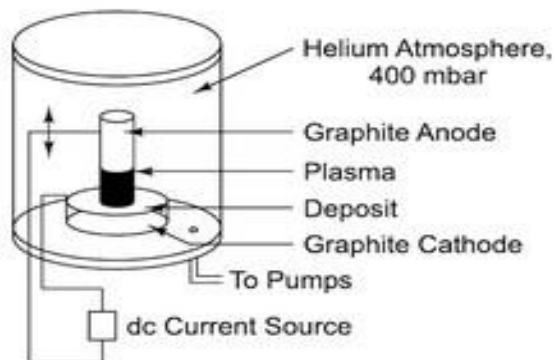
• STRIP OF A GRAPHENE SHEET ROLLED INTO A TUBE



Fabrications of Carbon nanotubes (CNTs)

1. Arc-evaporation Method or Plasma Arcing

The carbon arc discharge method is the most common and simple method. This method is also known as plasma arcing. In an enclosure, two carbon electrodes, separated by approximately 1 mm, are kept at a potential difference of about 20 V serving as anode and the cathode. The enclosure is filled with inert gas at low pressure. This experimental arrangement is shown in Fig. 5.18. A direct current of 50 to 100 A, driven by the external potential difference, creates a high temperature discharge between the two electrodes. The discharge vaporizes the surface of the positive electrode, and forms a small rod-shaped nanotube deposit on the negative electrode. As the tubes form the length, the positive electrode decreases and a carbon deposit forms on the negative electrode. When a small amount of cobalt, nickel or iron is added to the positive electrode, SWNTs are produced. If no catalyst is used the tubes are of MWNT type.



Important properties of CNT:

1. Stronger and most flexible molecular material because of C-C covalent bonding and hexagonal network architecture.
2. Young's modulus of over 1 Tpa vs 70 Gpa for Aluminum. 100 GPA for carbon fiber.
3. Strength to weight ratio 500 times greater than that for Al. similar improvements on steel and Titanium. One order of magnitude improvement over graphite.
4. Maximum strain is 10% much higher than any material.

5. Thermal conductivity is 3000W/mk in the axial direction with small values in the radial direction.
6. Electrical conductivity in order of magnitude higher than copper.
7. Can be metallic or semiconducting depending on chirality. Band gap can be adjusted. Electrical properties can be controlled through application of external magnetic field and mechanical force.
8. Very high current carrying capacity.

Medical/biomedical applications of carbon nanotubes:

Carbon nanotubes combined with biological systems can significantly improve medical science - especially diagnostics and disease treatment. Let us see few specific cases.

Anti-cancer treatment: When a patient goes through regular chemo therapy, chemotherapy doesn't destroy "bad" cells only. Along with those tumor cells, it destroys healthy cells too. He/she loses hair and this treatment has other side effects. With carbon nanotubes it is possible to destroy cancer cells alone without affecting normal cells. Scientists from Stanford University have discovered that nanotubes, when exposed to infrared light, tend to heat up to 160°F (70°C) in just 120 seconds. If they are placed inside the cancer cells, they simply destroy them. Such infrared radiation has no effect on normal cells.

Gene therapy: Gene therapy could also be improved by using carbon nanotubes. Let's say that a damaged or missing gene needs replacement from outside. But that's complicated - because DNA can't pass through the cell membrane. Modified carbon nanotubes can act as a transporter and change the gene.

Biomolecule sensor: Carbon nanotubes can be used for sensing the biomolecules or species. Many studies on the electrochemical reactivity of carbon nanotubes showed that carbon nanotubes promote electron transfer in heme containing proteins. Carbon nanotubes are able to access the heme centre of biomolecules that is generally not sensed by the glass electrodes.

Drug delivery: Carbon nanotubes can also be used as blood vessels in order to deliver drugs to their target. When the drug delivery is done that way, the drug dosage can be lowered. There are two methods, both equally effective - a) the drug can be attached to the side or behind, or b) the drug can actually be placed inside the nanotube for delivery.

Other biomedical applications: The ability to chemically modify the side- walls of carbon nanotubes also leads to biomedical applications such as vascular stents, and neuron growth and regeneration. Cells have been shown to grow on carbon nanotubes, so they appear to have no toxic effect. The cells also do not adhere to the carbon nanotubes, potentially giving rise to applications such as coatings for prosthetics and anti-fouling coatings for ships.